PHYSIOLOGIC STATE CLASSIFICATION AND INTERPRETATION FRAMEWORK

Physiologic State Interpretation Concept

The longitudinal penile physiology monitoring device may classify physiologic conditions through computational interpretation of multi-modal sensing data. Rather than reporting raw sensor outputs alone, the system may identify higher-order physiologic states representing functional biological conditions inferred from correlated mechanical, vascular, thermal, and motion-related measurements collected over time.

Physiologic state classification may operate continuously or intermittently and may adapt based on individualized baseline characteristics learned through longitudinal monitoring.

Example Physiologic State Categories

Non-limiting physiologic states may include flaccid state, transitional tumescence, partial rigidity, stabilized erection, recovery phase, nocturnal activation state, resting baseline state, and post-event recovery behavior. States may be represented as discrete categories, continuous indices, probability distributions, or multi-dimensional physiologic descriptors.

Multi-Modal Classification Inputs

Classification algorithms may utilize combined inputs including deformation measurements, mechanical compliance data, perfusion signals, temperature trends, inertial context, temporal patterns, and historical behavioral information. Weighted sensor fusion may improve robustness against noise, device repositioning, or environmental variability.

Interpretation may incorporate rule-based logic, statistical modeling, probabilistic inference, pattern recognition techniques, or machine learning frameworks capable of identifying physiologic patterns across diverse user populations.

Temporal and Longitudinal Interpretation

Physiologic state classification may consider temporal continuity rather than isolated sensor events. Transition timing, persistence of state, recurrence frequency, and stability duration may contribute to interpretation outputs. Longitudinal analysis may enable detection of trends, adaptation patterns, or deviations from individualized baselines.

Context-Aware Interpretation

Contextual information derived from motion sensing, sleep-state detection, environmental conditions, or user-selected operational modes may modify interpretation weighting. Context-aware processing may reduce false positives, improve classification confidence, and allow physiologic states to be interpreted relative to behavioral conditions.

Output Representation and Abstraction

Classified physiologic states may be presented through abstract representations such as indices, scores, symbolic indicators, graphical trends, or non-explicit visualizations designed to convey physiologic insight without anatomical imagery. Output abstraction may support privacy-preserving data representation and cross-platform interoperability.

The classification frameworks described herein are illustrative and non-limiting and encompass present and future computational intelligence methods capable of interpreting wearable-derived physiologic data into meaningful state representations.